Activity 1: Business Capability Decomposition

1. Core Business Domains:

- User Management
- Catalog Management
- Fulfillment
- Inventory
- Pre-purchase (Shopping Cart)

2. Microservices Definition (Result Table):

Business Capability	Proposed Microservice	Owned Data (Entities)
User Management	**User Service**	User Profile, Auth Credentials.
Catalog Management	**Product Service**	Product Details, Price, Image Links.
Fulfillment & Tracking	**Order Service**	Order Information (Header), Line Item Details, Shipping Address.
Inventory	**Inventory Service**	Stock Quantity, Warehouse Location.
Pre-Purchase	**Cart Service**	Cart Contents, Session Status.

3. External Dependencies: ShopSphere needs to communicate with the following third-party systems:

- **Payment Gateway:** (E.g., Stripe, PayPal, VNPay) for payment processing.
- **Email/SMS Provider:** (E.g., SendGrid, Twilio) for sending order confirmations/OTPs.
- **Shipping Carrier API:** (E.g., GiaoHangNhanh, FedEx) for calculating shipping fees and obtaining tracking numbers.

Activity 2: Service Contracts Definition

1. API Definition for Product Service:

This is the RESTful API documentation for the product service.

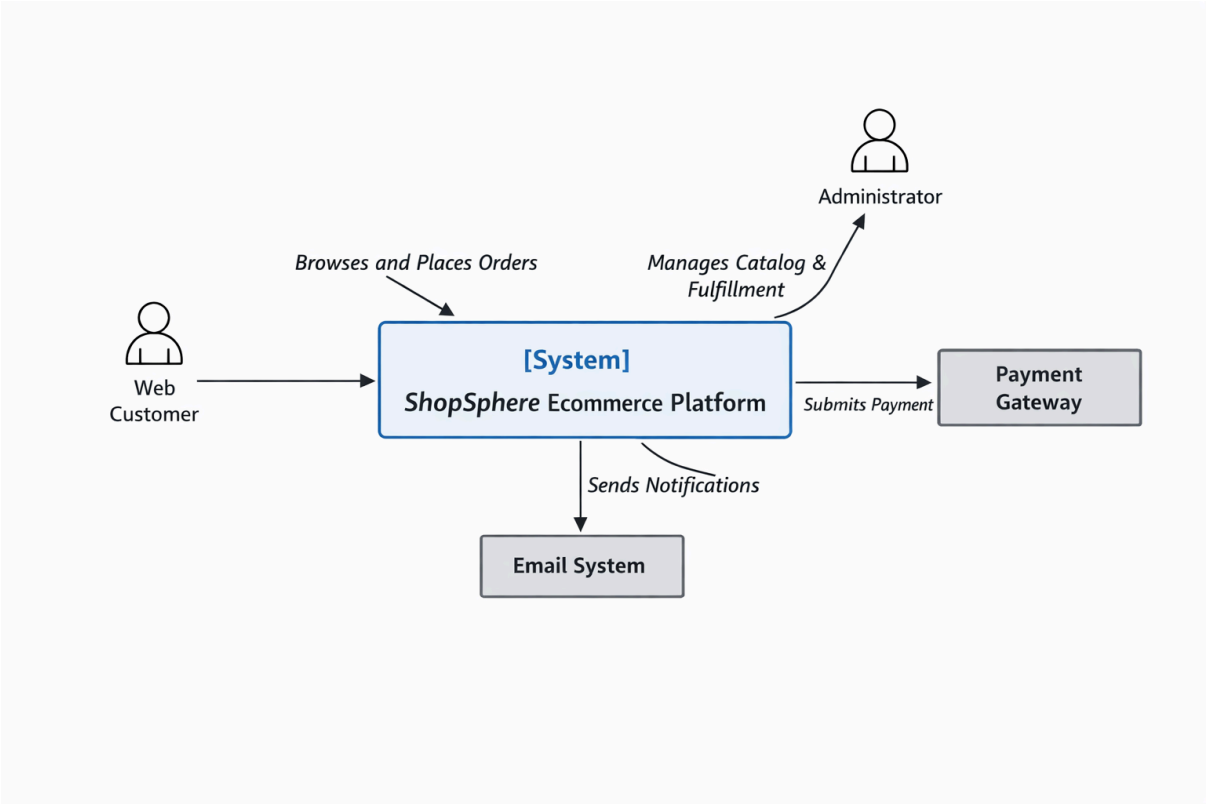
Endpoint	HTTP Method	Description	Return Data
/api/products/{id}	**GET**	Retrieve full details of a product.	Product Object (name, price, description,

			stock_id)
/api/products	**GET**	Search/List products based on criteria.	List of Products (summary form)
/api/products	**POST**	Add a new product (For Admin).	Newly created Product Object

2. Cart Service Interaction:

- **Problem:** When a user adds an item to the cart, the Cart Service needs to know the product's Name and Price.
- **Rule:** The Cart Service **ABSOLUTELY MUST NOT** directly access the Product Service Database.
- **Solution:** The Cart Service will send an HTTP Request (GET) to the /api/products/{id} endpoint of the Product Service to retrieve the data.

Activity 3: C4 Model (Level 1: System Context)



4. Communication Strategy Analysis:

At this level, we define when to use Synchronous (awaiting immediate response) and Asynchronous (send and forget/process later) communication.

Interaction	Between Which Services?	Communication Type	Rationale

Customer Login / Lookup	API Gateway -> User Service	**Synchronous (…**	The user needs an immediate response to log in or view their profile. They cannot be made to wait.
Order Confirmation Email	Order Service -> Email Provider	**Asynchronous…**	Sending an email is time-consuming. It's unnecessary to make the user wait for the email to be sent before reporting "Order successful". The system will accept the order first, and the email is sent later (fire-and-forget).
Inventory Check	Cart Service -> Inventory Service	**Synchronous (…**	Stock must be verified immediately before allowing payment (Checkout) to prevent overselling.